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Improvement of Anaerobic Digestion of Landfill Leachate by Using Coagulation-Flocculation, Fenton's Oxidation and Air Stripping Pretreatments

Yosr Smaoui,[a] **Najwa Mlaik**,[b] **Jalel Bouzid**,[a] **and Sami Sayadi**[b]

[a]Laboratoire Génie de l'Environnement et Ecotechnologie, Ecole Nationale d'Ingénieurs de Sfax, Université de Sfax-Tunisie, PB 1173, Sfax, 3038, Tunisia; smaoui_yosr@yahoo.fr (for correspondence)

[b]Pôle d'excellence régionale AUF, (PER-LBP), Centre de Biotechnologie de Sfax, BP 1177, Laboratoire des Bioprocédés Environnementaux, Sfax, 3061, Tunisia

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In Tunisia, the anaerobic digestion process is regarded as a new alternative for the treatment of landfill leachate (LFL). Raw LFL contained high amounts of organic matter, high ammonia concentrations and low biodegradability. In order to improve anaerobic conditions and biogas yields, pretreatment using coagulation-flocculation, Fenton oxidation (FO) and air stripping processes were investigated and compared in terms of biogas production. Air stripping process was optimized using Response Surface Methodology (RSM). Central Composite Design (CCD) was used with three factors. Under optimized conditions, removals of 85% ammonia, 26% chemical oxygen demand (COD) were achieved. Coagulation-flocculation provided an increase in the effluent biodegradability with 46% of COD reduction. Fenton's oxidation allowed reducing 48% COD, 30% of biological oxygen demand (BOD_5). The effect of pretreatments was investigated with the biochemical methane potential (BMP) assay. A comparison of three different pretreatment methods of LFL for biogas production was performed. Air stripping was demonstrated as providing a better biogas yield than other pretreatments which reached 588 mL/g CODint. The best methane yield was obtained using air stripping pretreatment under optimized conditions. It can result in more robust and effective anaerobic digestion. © 2017 American Institute of Chemical Engineers Environ Prog, 00: 000–000, 2017

Keywords: landfill leachate, coagulation-flocculation, fenton oxidation, air stripping, methane yield

INTRODUCTION

Landfill leachate is a contaminated liquid formed by excess water percolating through the waste layers in landfills. It contains a high concentration of organic contaminants, ammonia, salts, even heavy metals and xenobiotic organic compounds such as phthalates [1] that can cause severe environmental impacts by contaminating surface and ground water [2]. To minimize these impacts, efficient, environmentally and economically viable processes for the landfill leachate treatment are required.

In Tunisia, treatment of leachate stations adopts a strategy consisting of a set of successive operations that are coagulation-flocculation followed by an aerobic membrane bioreactor, ultrafiltration and finally a polishing step by reverse osmosis membranes. Tunisian LFL is characterized by a high concentration of chemical oxygen demand (COD> 30 g/L) and ammonia nitrogen (>3 g/L), in addition to a low biodegradability and the presence of some heavy metals. Moreover, the high concentration of organic matter present in Tunisian landfill leachate, make the treatment process costly in terms of the frequent use of chemicals and the high energy requirements.

The main idea of this study was to propose an alternative approach for the treatment of a leachate generated in a controlled discharge in Tunisia. Anaerobic digestion (AD), the most popular biological treatment, produces moderate amounts of sludge, quite effective for young leachate and generates energy from the methane produced making this process economically and environmentally advantageous [3,4].

Various studies showed the presence of a wide variety of substances that are considered as inhibitors of the anaerobic digestion process [5]. However, an obstacle for biological treatment of leachate is the high ammonia concentration that can inhibit the growth of microorganisms and deteriorate the performance of the treatment. It was suggested that inhibition might occur in the range of 1500–7000 mg/L of total ammonia nitrogen [6]. The C/N ratio is considered as a key factor for anaerobic digestion. In fact, if this ratio is small, which is the case of some leachate, nitrogen is liberated, accumulates in the form of ammonia and begins to apply a toxic effect when the pH is higher than 8.5 [7]. Therefore, the low C/N ratio and the poor biodegradability of landfill leachate reduce the effectiveness of biological treatment processes and require preliminary physico-chemical treatments [8].

As already mentioned, the coagulation-flocculation process, is the first treatment used in the leachate treatment plants in controlled discharges of Tunisia. This process is widely used because of its easy operation, low capital and operating cost and environmental impact [9,10].

Various studies have shown the effectiveness of this process for pretreatment of wastewater [11,12], which reduces pollutant loads, removes recalcitrant compounds and therefore, improves the biodegradability of wastewater during the biological treatment. Selection of appropriate coagulant, the

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Table 1. Main characteristics of LFL collected during the study period (average of three samples).

Characteristics	Unit	LFL value
pH		8 ± 0.2
Conductivity	mS/cm	36.8 ± 2.8
Total COD	mg/L	$40{,}000 \pm 3600$
Soluble COD	mg/L	$36{,}600 \pm 600$
BOD_5	mg/L	$11{,}000 \pm 1250$
TKN	mg/L	3250 ± 250
NH_4-N	mg/L	2760 ± 50
Pb	mg/L	< 0.005
Cd	mg/L	< 0.005
Zn	mg/L	< 0.005
Fe	mg/L	11 ± 1.8
Cr	mg/L	1.7 ± 0.07
Ni	mg/L	0.05 ± 0.05

effect of pH and investigation of the dosage of optimal reagents are required to reach optimal performance.

Advanced Oxidation Processes (AOPs) such as ozonation, UV/H_2O_2, photocatalytic oxidation are considered clean technologies for the treatment of wastewater [13]. The highly oxidative hydroxyl radical ($OH^{\bullet}$) produced in this process can oxidize refractory organic matter into more biodegradable compounds, then improves the biodegradability of effluent that can be further treated by an economical biological process [14,15]. Therefore, Fenton process, which is one of the AOPs, is technologically simple, iron and hydrogen peroxide used are not toxic and can enhance the biodegradability of wastewater; further facilitating the following biological process which may be more reasonable so as to avoid high energy consumption [16]. Ammonia or air stripping is the most widely employed treatment for the removal of NH_3-N from landfill leachate [17]. The concentrated gaseous ammonia purged after stripping can be recovered and absorbed by strong acidic solutions, such as sulfuric acid (H_2SO_4), forming a mineral fertilizer for agricultural use [2]. To reduce the cost of the process and to select the most economical parameters (pH, aeration rate and reaction time) that offer a suitable C/N ratio for the subsequent anaerobic, optimization of the pretreatment can be a vital practice to improve the total efficiency of the process. Response surface method (RSM) is a statistical tool for optimization of different processes for wastewater treatment. In this study, a central composite design (CCD) was used to optimize the three factors, pH, aeration rate and reaction time as the main factors in the air stripping process.

The main objective of the present study was to investigate the effect of coagulation-flocculation, FO with zero valent iron and air stripping on biodegradability and C/N ratio improvement. The effect of these pretreatments processes on the anaerobic digestion was investigated.

MATERIALS AND METHODS

Landfill Leachate Sampling and Inocula

The leachate used in this study was obtained from a discharge in the Sfax region of Tunisia. It is taken straight from the collection box, and kept at 4°C in the dark before being used in the experiments to avoid its decomposition. The mean physico-chemical characteristics of raw leachate are shown in Table 1. For batch anaerobic digestion experiments, the seed sludge was taken from an anaerobic digester installed in a WWTP located in CHARGUIA, Tunisia.

Pretreatment Experiments

Coagulation-Flocculation Process

Coagulation-flocculation experiments were carried at room temperature using a Floclab Jar Test (Prolabo-France) equipped with six 500 mL beakers volume. The chemical reagents used were ferric chloride ($FeCl_3$.$7H_2O$) as coagulant and polyacrylamide as a flocculant. A 500 mL leachate sample was poured into each jar. Then, the pH was adjusted to the desired value using 1 M HCl or 5 M NaOH. Hydrochloride acid, sodium hydroxide and hydrogen peroxide were purchased from Merck (France). The experimental process consisted of three subsequent phases according to Smaoui et al. [18]. A flash mixing started at 150 rpm and lasted 4.5 min during the coagulant addition. The flocculation process was conducted for 60 min by slow stirring at a speed of 48 rpm. After that, a decantation time of 30 min was performed. Samples were taken from the liquid level about 3 cm below the surface for chemical analyses.

Fenton Oxidation Process

The iron metal used in this study was obtained from a metal turner and was added to the leachate solution. Then, hydrogen peroxide was introduced to start the Fenton reaction while a continuous stirring at 200 rpm was applied. The chain reaction in the Fenton process is shown in Eqs. 1–5 [19].

$$Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH^{\bullet} + OH^{-} \quad (1)$$

$$OH^{\bullet} + Fe^{2+} \rightarrow Fe^{3+} + OH^{-} \quad (2)$$

$$Fe^{3+} + H_2O_2 \rightarrow Fe^{2+} + HOO^{\bullet} + H^{+} \quad (3)$$

$$OH^{-} + H_2O_2 \rightarrow HO_2^{\bullet} + H^{+} \quad (4)$$

$$H_2O_2 + OH \rightarrow H_2O + HO_2 \quad (5)$$

But, in the case of the use of zero-valent iron in place of ferrous salts, a natural corrosion of the iron sheet surfaces would occur in the presence of hydrogen peroxide leading to the production of iron species in acidic conditions according to Eq. 6 [20].

$$Fe^0 + 2H^{+} \rightarrow Fe^{2+} + H_2 \quad (6)$$

To finish the production of OH radicals and to remove residual hydrogen peroxide, pH of the samples were brought to more than 10 using 10 M NaOH [21].

Air Stripping Process

Design of Experiments

The air stripping process was designed and optimized using RSM. The experimental design and optimization were conducted with NEMROD W software. A Central Composite Design (CCD) was used for three variables and three levels, to optimize three factors i.e. pH, aeration rate and reaction time as the main factors of the air stripping process. The ranges and the levels of the variables investigated in this study are given in Table 2. The range of pH values selected was based on the best performance of the application. The aeration rates were chosen to avoid evaporation problems. The reaction time was chosen on the basis of preliminary experiments. The response of the variable (ammonia removal) was fitted to a second-order model that describes the relationship between independent variables and the dependent output variables (y) and to reveal the interaction

Table 2. Experimental field.

	Factor	Unit	Center	Variation path
X1	pH		10.500	1.500
X2	Aeration rate	L/min	4.500	2.500
X3	Reaction time	h	12.000	6.000

between the factors. The better model was developed by a second-order mathematical model (Eq. 7).

$$\hat{Y} = b_0 + \sum_{J=1}^{4} b_J X_J + \sum_{J=1}^{4} b_{JJ} X_J{}^2 + \sum_{\substack{k=1 \\ J \neq k}}^{4} b_{Jk} X_J X_k \quad (7)$$

Where $\hat{Y}$ is the predicted response, b_0 is the model constant, b_J is the linear effect, b_{JJ} is the squared effect, b_{Jk} is the interaction effect and X_J are independent variables. The ridge analysis was employed to optimize the experimental condition and allow higher ammonia removal. The quality of the fit polynomial model was expressed by the coefficient of determination R^2 and the statistical significance was checked by Fisher's *F*-test. Three-dimensional plots with the respective contour plots were obtained from the results of the experiments.

Air Stripping Experiments

Two liters of LFL were pured in the air stripping unit, which consisted of a Plexiglas column. The experimental conditions are indicated in Table 3. Air charged with ammonia was released through the top of the stripping column. The (NH_4-N) supernatant was measured and the optimal conditions for the removal of most ammonia determined.

Batch Anaerobic Digestion

Batch anaerobic digestion was conducted to further examine the effect of each pretreatment studied on biogas production. These experiments were carried out under mesophilic conditions (37 ± 1°C) for 50 days, using 500 mL glass bottles as reactors. Each bottle was fed with an appropriate amount of substrate and inocula, keeping a volatile solids (VS) ratio (VSs to VSi) at 1:1 in all setups [22]. Five experiments bottles were performed. They contain: anaerobic sludge without LFL, anaerobic sludge with raw LFL, anaerobic sludge with LFL treated by coagulation-flocculation, anaerobic sludge with LFL treated by FO and anaerobic sludge with LFL treated by air stripping. The control batch was conducted only with inocula, to estimate the volume of biogas resulting from the auto-fermentation of the inocula biomass. The pH of the leachate samples was adjusted to 7.0. All bottles were purged with a gas mixture of 75% N_2 and 25% CO_2 for 3–4 min to supply anaerobic conditions. The volume of the produced biogas was monitored daily by the displacement method, and the water and corresponding biogas volumes were calculated. The direct measurement of the methane produced was carried out using an alkaline solution 5% (w/v) instead of water, which effectively absorbs all the content of CO_2 in the off-gas [23]. All results are presented as averages of duplicates with standard deviation.

Analytical Methods

The physico-chemical analysis of leachate samples was performed according to American Standard Methods for Water and Wastewater Examination [24].

The pH of the solution was determined using a pH meter (Metrohm). The electrical conductivity was measured by WTW Multiline 3410 devices. The chemical oxygen demand (COD) values were determined by titration after a total digestion with H_2SO_4 and potassium dichromate at 150°C for 2h [2]. The biological oxygen demand (BOD_5) value was determined by the respirometric method using WTW OxiTop. The total Kjeldahl nitrogen content (TKN) and the ammoniacal nitrogen (NH_4-N) were analyzed as a Kjeldahl-N method. The total concentrations of Fe, Ni, Cr, Pb, Cd, and Zn were determined using atomic absorption flame emission spectroscopy AAS (Thermo Scientific).

Table 3. Central composite design, experimental conditions, and measured responses.

N° Experience	pH	Aeration rate	Reaction time	Ammonia removal (%)
1	−1	−1	−1	26.00
2	1	−1	−1	29.50
3	−1	1	−1	22.87
4	1	1	−1	71.70
5	−1	−1	1	33.40
6	1	−1	1	63.00
7	−1	1	1	48.25
8	1	1	1	95.20
9	−1	0	0	34.00
10	1	0	0	68.23
11	0	−1	0	41.48
12	0	1	0	70.40
13	0	0	−1	52.52
14	0	0	1	73.40
15	0	0	0	59.00
16	0	0	0	65.00
17	0	0	0	60.00

Statistical Analysis

All the results were expressed as a mean ± standard deviation, and the statistical evaluation was performed using one-way ANOVA (analysis of variance) method followed by post hoc Tukey's test with a 0.05 significance level to determine whether the different pretreatments were statistically different from each other or not.

RESULTS AND DISCUSSION

Physico-Chemical Characterization

As a first step, it is important to determine the physico-chemical characteristics of the leachate used in this study. The analytical data are presented in Table 1. The leachate samples are characterized by alkaline pH ranging from 8 to 8.2 with a dark brown color, which indicate the high presence of humic substances [25], a high electrical conductivity (EC = 36.8 mS/cm), a high organic load expressed in COD with a mean value of 40 g/L and a BOD_5/COD ratio of 0.27 indicating the low biodegradability of the leachate. The Nitrogen concentration was also high mainly in ammonium. This is correlated with what was reported in the literature [26]. The NH_4-N present in the solution is an essential nutrient for anaerobic bacteria, but it became toxic and constituted an inhibitor for the anaerobic treatment at high concentration [27,28]. It's apparent from Table 1 that leachate metal contents are low; nevertheless, some micro-pollutant minerals such as iron and chrome are also present.

Coagulation-Flocculation Pretreatment

The first step was to search for the optimum pH of the coagulation-flocculation process. Therefore an optimization was carried out by following the evolution of COD removal at different pH ranging from 3 to 7 (Figure 1a). It was observed that COD removal from leachate at pH 7 was 35%. However, an improvement of the results was obtained by

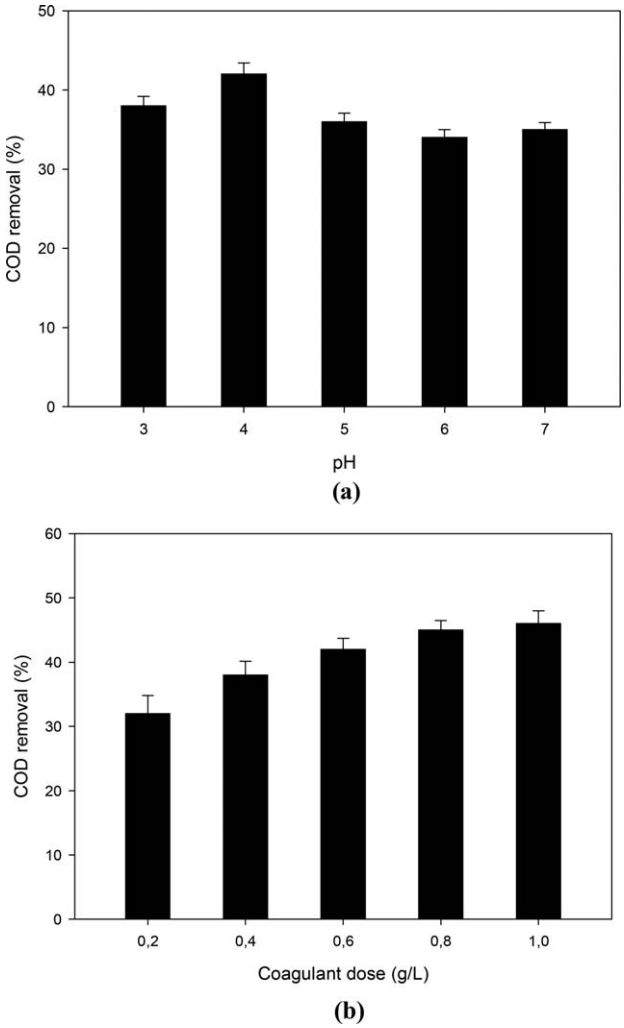


Figure 1. Variations in COD Removal of LFL during coagulation-flocculation pretreatment: effect of pH (a) (Coagulant dose: 0.6 g/L, flocculant dose: 20 mg/L); effect of coagulant dose (b) (Initial pH: 4, flocculant dose: 20 mg/L).

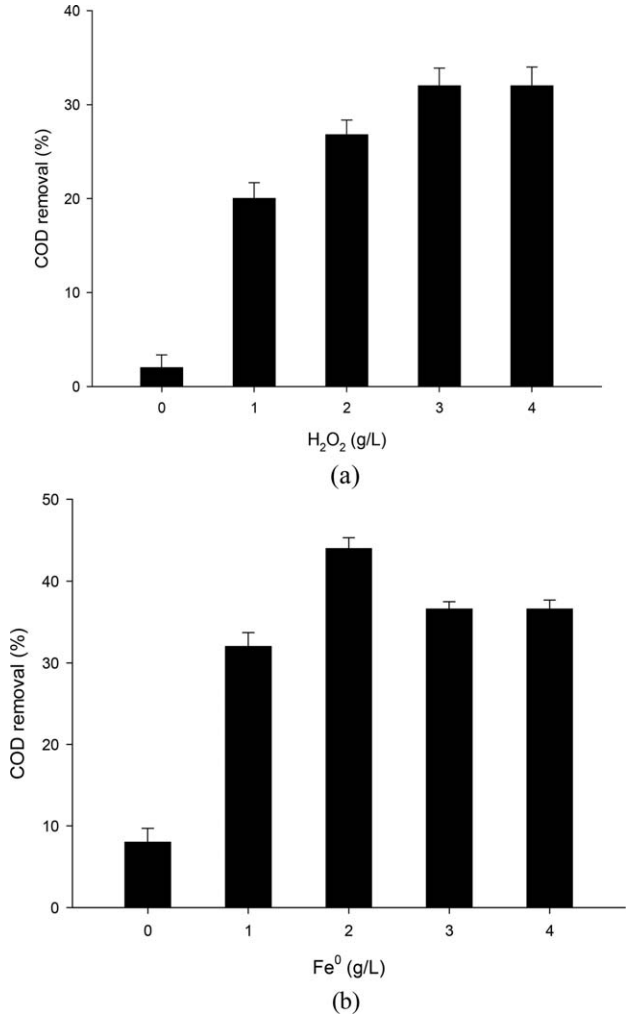


Figure 2. Variations in COD removal of LFL during Fenton oxidation pretreatment: effect of H_2O_2 (a) (Initial pH: 3, Fe^0: 1 g/L, reaction time: 2 h), effect of Fe^0 (b) (Initial pH: 3, H_2O_2: 3 g/L, reaction time: 2 h).

decreasing the pH value and the most significant COD removal occurred at pH 4. In fact, the increase of the solution pH resulted in a reduction in the retention of ferric chloride particles. This can be explained by the pH effect on the nature of produced polymeric metal species that will be formed as soon as the metal coagulants are dissolved in water. According to Li et al. [29], the removal efficiency is better at acidic than the basic condition and this can be due to the different hydrolyzed species of ferric ion. Maranon et al. [30] suggested that pH 3.8 was considered as the optimum condition for this process. On the other hand, Guerreiro et al. [4] showed that the best removal of COD and BOD_5 was obtained at pH 3. At this pH, the coagulation-flocculation occurs probably by a mechanism of charge neutralization [31]. In fact, we note that the extent of pH range is affected by different factors, such as the types of coagulant used, the chemical composition of wastewater and the concentration of coagulant. To optimize the COD removal, the concentration of Fe^{2+} was progressively varied from 0.2 to 1 g/L at pH 4 (Figure 1b). The optimal COD removal of 46% was obtained at a concentration of Fe^0 equal to 0.8 g/L. Above this dose, no further COD removal was observed. The overdose of coagulant resulted in the absorbance of the cations by colloids, which become positively charged and led to an electronic repulsion and decreased the capacity of organic matter removal [11]. Others studies have proven that the addition of coagulants such as Fe^{2+} resulted in the increase of COD removal which is associated with the decrease of humic substances [32].

Optimization of the Fenton Process

An optimum oxidation can be obtained if the consumption of hydroxyl radicals by the reagents is limited. This implies that neither hydrogen peroxide nor ferrous ions should overdose, so the ratio $[H_2O_2]/[Fe^{2+}]$ should be optimized to obtain the maximum amount of hydroxyl radicals and after that oxidize organic pollutants. Experiments involving H_2O_2 doses within the range of 0 - 4 g/L were carried out at pH 3 using 1 g/L of Fe^0. As shown (Figure 2a), the increase of H_2O_2 concentration had a positive effect on COD removal. Thus, optimal COD removal of 30% was obtained at 3 g/L H_2O_2. The further increase of H_2O_2 concentration did not increase the COD removal efficiency. This can be explained by the saturation of the oxidation process [33]. This result was also found in other studies which explained this phenomenon by the auto-decomposition of H_2O_2 to oxygen and water, and the recombination of $OH^\bullet$ radicals [34]. The COD removal increased gradually with increasing Fe^0 and reached 44% at 2 g/L of metallic

Table 4. Variance analysis for CCD design.

Source of variation	Sum of square (SS)	Degrees of freedom (DF)	Mean square	Rapport	Sig
Regression	6346.65	9	705.18	34.3700	< 0.01***
Residual	143.62	7	20.51		
Lack of fit	122.95	5	24.59	2.3798	32.2 NS
Pur error	20.66	2	10.33		
Total	6490.2	16			

***Indicates significant at the level of 99.99%; NS: Indicates nonsignificant at the level 95%.

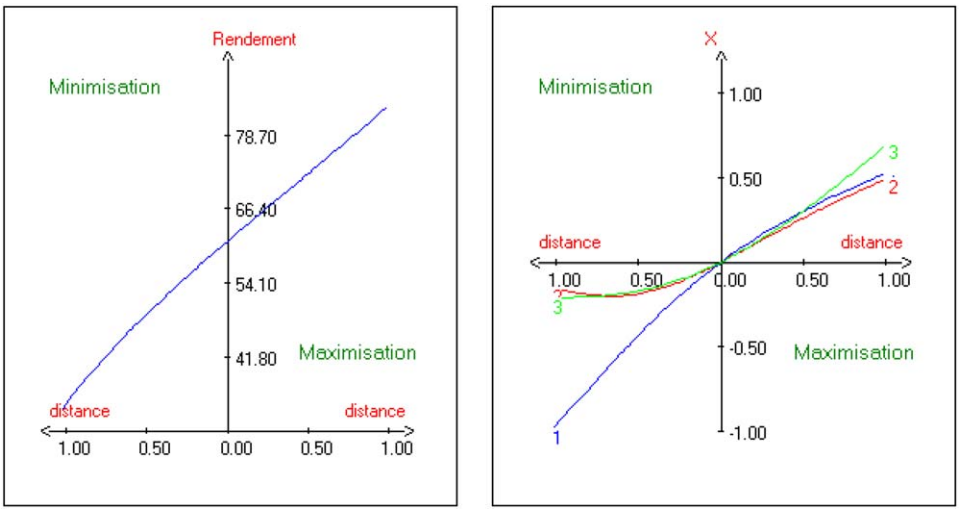


Figure 3. Ridge analysis: optimal path plot (a) and optimal coordinate plot (b). [Color figure can be viewed at **wileyonlinelibrary.com**]

iron and 3 g/L of hydrogen peroxide (Figure 2b). Beyond this concentration, COD removal decreased and remained steady. From the reaction, the degradation of organic matter is slow because of the small production of iron II. After that, the concentration of iron increased interactions with hydroxyl radicals, which causes the maximum removal of COD and the dark color becomes brown. The overdose of Fe^{2+} or H_2O_2 can react with the hydroxyl radical and inhibits the oxidation reaction [20]. In comparison with other studies, Mohajeri et al. [34] were observed a COD reduction of 58% with H_2O_2 and Fe^{2+} concentrations of 0.033 mol/L and 0.011 mol/L, respectively. Lopez et al. [35] observed a 60% COD removal by using reagent dosages as high as 10,000 mg/L of H_2O_2 and 830 mg/L of Fe^{2+} within 2 h. Boumechour et al. [36] proved that the maximum COD removal (64%) was achieved by using the optimum operational conditions of 4 g/L H_2O_2 and 1 g/L Fe^{2+}.

Optimization of the Air Stripping Process

Model Equation and Validation

The optimization of the experimental conditions of ammonia removal was achieved using a CCD design and carried out under several experimental conditions including pH, aeration rate and reaction time. The results of the variance analysis for the fitted model are summarized in Table 3. The regression sum of square is statistically significant at the level 99.9%. Moreover, the coefficient of determination R^2 was 0.978; this means that 97.8% of the observed variations of the response are attributed to the variable effects, and the experimental values are in excellent agreement with the predicted values of the reply. This reveals the high validity and the proper fitting of the model obtained for ammonia removal. On the other hand, a comparison between the variance related to the lack of fit to that of pure error has been established and the results show the nonsignificance of the lack of fit (Table 4). The model obtained was used to predict response values in the domain studied and to draw isoresponse contour plots and response surfaces.

Interpretation of the Response Surface Model

Ridge Analysis

The representation of the optimum path for the analysis of the removal of the ammonia by the response surface is shown in Figure 3. The method consists in plotting the curves in order to determine the maximum values of the response. The left and right parts of both plots indicate the minimization and the maximization of the response, respectively. Figure 3a shows that the maximum ammonia removal increased from the center to the boundary. Figure 3b displays the coordinates for each factor, in codified variables. It can be seen that maximum ammonia removal is obtained by adjustment of all the factors to a high value.

Response Surface and Contour Plot

The response surface and contour plots are the most revealing ways of illustrating and interpreting the response surface system. They are used to evaluate the influence of

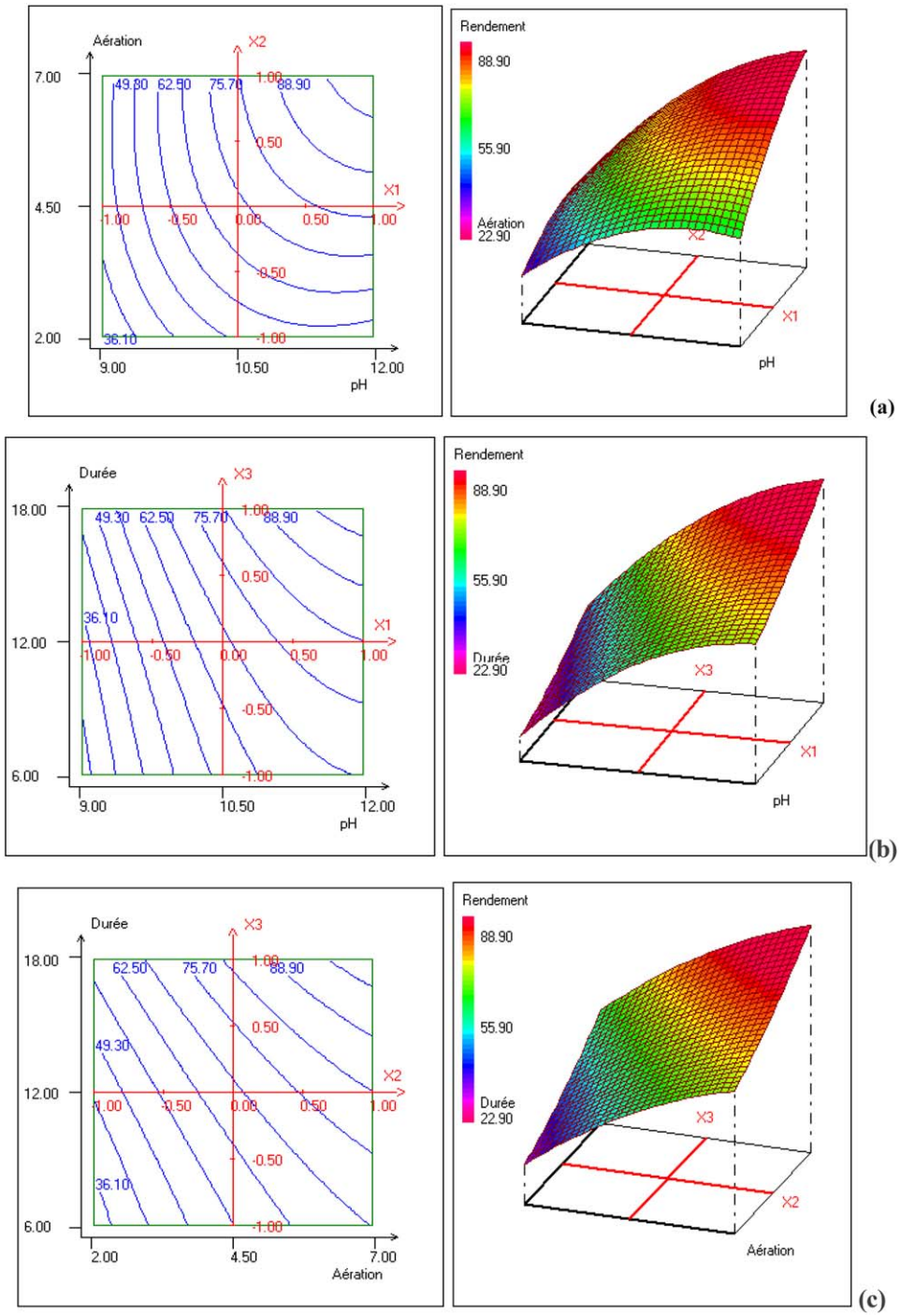


Figure 4. Three-dimensional response surface and contour plots for the effect on ammonia removal: pH and aeration rate at constant time 18h (a), pH and time contact at constant aeration rate 7L/min (b), aeration rate and time at constant pH 12 (c). [Color figure can be viewed at **wileyonlinelibrary.com**]

pH, aeration rate and the duration of the reaction on ammonia removal by the air stripping process. This method consists in plotting the response variation against two factors while the third is held constant (Figure 4).

Figure 4a shows the high interaction between pH and aeration rate. Indeed, ammonia removal increased with the simultaneous rising of both factors. However, when the pH was less than 10.5, the phenomenon was primarily governed by the rate of the aeration factor. Beyond this value, the pH became predominant. To maximize the response, we must increase both the pH and the aeration rate. Figure 4b represented the isocontour plot when the level of the aeration rate was fixed at 7 L/min. It's clear that yields higher than 85% could be reached by fixing the pH at 12 and reaction time at 18 h. These results are in agreement with those of Guo et al. [37] who obtained a removal of 96.6% of ammonia at pH 11.0 and an aeration time of 18 h. Similarly, Marttinen et al. [25] have shown a removal of 89% at pH 11 and aeration time of 24h. Finally, it can be seen in Figure 4c, that a higher level of ammonia removal was obtained with high levels of both aeration rate and reaction time. Amokrane et al. [38] have proved that the high level of NH_3

is the most important problem faced by the landfill operator that reduces the performance efficiency treatment, and the increase in ammonia removal is caused by the increase in aeration rate and contact time. The objective of this optimization was to obtain removal of 85% of ammonia to improve the C/N ratio. The conditions are pH 11, aeration rate of 7 L/min and 18 h reaction time. These conditions have been confirmed experimentally and allowed the removal of 85% ammonia.

Table 5. COD, BOD_5 and ammonia reductions obtained in each treatment step.

	COD reduction (%)	BOD_5 reduction (%)	Ammonia reduction (%)	BOD_5/COD
Coagulation-flocculation	46	30	18	0.37
Fenton's oxidation	48	30	23	0.39
Air stripping	26	18	85	0.32

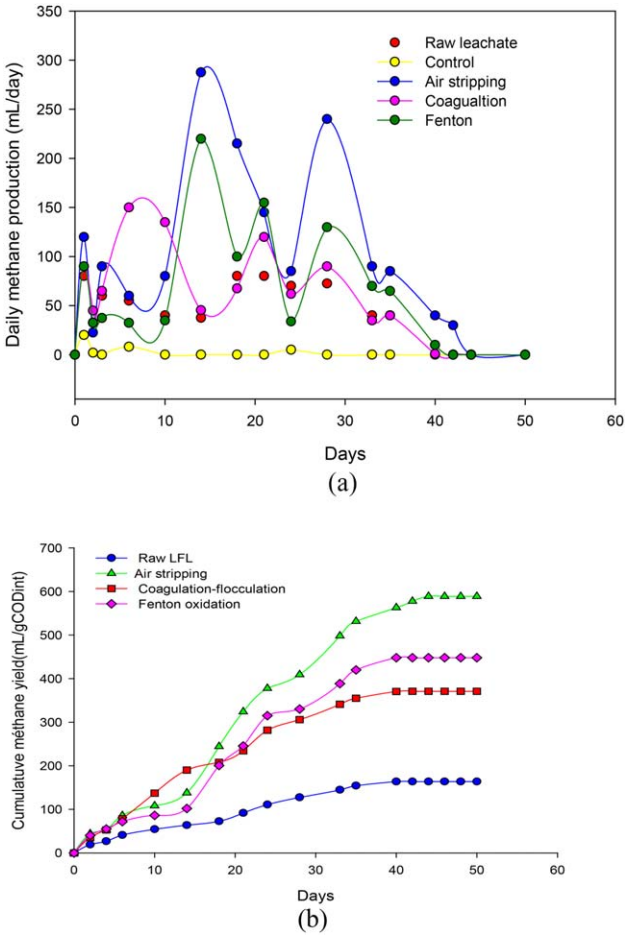


Figure 5. Daily methane production rates (a) and cumulative methane yields (b) during anaerobic treatment. [Color figure can be viewed at **wileyonlinelibrary.com**]

COD, BOD, and NH_4 Reductions Obtained in Each Pretreatment Step

The COD, BOD and ammonia reduction obtained after the coagulation-flocculation, Fenton's oxidation and air stripping pretreatments are presented in Table 5. An improvement of leachate biodegradability after all the pretreatments applied was observed. Therefore, the BOD/COD ratio increased from 0.28 to 0.32, to 0.37 and 0.39 for air stripping, coagulation- flocculation and FO respectively.

Therefore, FO and coagulation- flocculation exhibited a high performance to remove organic matter and thus, to enhance the biodegradability of the effluent. In addition, the zero valent iron used during FO treatment is cheap and non-toxic. In spite of the mentioned advantages, these two techniques present some drawbacks like the low ammonia reduction (Table 5), low pH requirement and the generation of sludge [19]. After, applying the tested pretreatment process, LFL can be classified as a biodegradable effluent and, thus, the anaerobic digestion can be carried out [38]. These results are in agreement with Ghanbari et al. [39] who showed a biodegradability improvement from 0.11 to 0.24 and 0.4 after pretreatment of LFL with coagulation and Fenton process. Lopez et al. [35] observed that Fenton process could be useful to improve the biodegradability of leachate that can be treated in a subsequent biological reactor. Vilar et al. [40] have shown a removal of organic matter and an increase of the biodegradability after FO of leachate. Compared to other treatments, the air stripping leads to high ammonia removal, which is an inhibitor of the anaerobic digestion [7], and therefore improves the C/N ratio (Table 5). Nevertheless, this process presents the necessity of a high pH value [41].

Anaerobic Digestion on Batch

The daily methane production and cumulative methane yields of different landfill leachate samples are presented in Figure 5. A batch control was also monitored and contained the only inoculum. The biogas production run for about 50 days until no more biogas production was observed. Figure 5a showed that methane production started immediately on the first day in all batches. For raw LFL, the highest methane production was obtained during 20 days with peaks of 80 mL at day 18. After 30 days, a progressive decrease of biogas production was observed until the 50th day when no methane was produced. It can be seen that air stripping pretreatment offers the most significant daily methane production with a peak of 287 mL on the 14th day. A second peak was observed on the 28th day with 240 mL. After that, a progressive decrease of biogas production was observed. For the coagulation and FO pretreatments, the most usual daily biogas observed was 150 and 220 mL at days 6 and 14,

Table 6. Characteristics of landfill leachate after anaerobic treatment.

	Raw	Pretreated by Coagulation-flocculation	Pretreated by Fenton's oxidation	Pretreated by Air stripping
pH	7.93	7.96	8.09	8.29
COD removal (%)	68	75	77	85
Biogaz yield (mL/g CODint)	163.69	370.9	448	588.88

respectively. After that, a decrease of methane production was noticed. Figure 5b gives the cumulative methane yield during batch anaerobic digestion of raw and pretreated LFL. It's clear that the methane yield was improved for all the pretreated leachate when compared to raw LFL (raw LFL- air stripping: $P = 0.000$; raw LFL – coagulation-flocculation = 0.003; raw LFL-Fenton: $P = 0.000$). At the end of digestion, the methane yield was 0.163.7 L/g CODint in raw LFL, which is lower than values reported in other reports [5,42]. Indeed, Chen et al. [5] showed that methane production rates of 0.35 L/g CODint during anaerobic digestion under optimal condition. Whereas, Kennedy and Lentz [42] reported a methane gas generation rate of 0.29–0.34 L/g CODint for the treatment of landfill leachate in up flow anaerobic sludge blanket reactors. Methane yields were of 370.9 mL/g CODint, 448.6 mL/g CODint and 588.9 mL/g CODint in coagulation-flocculation, FO and air stripping pretreatment, respectively. Obviously, a classification of the pretreatments used, laid out by increased methane yield, is as follows: "untreated leachate < coagulation-flocculation < Fenton oxidation < air stripping. The improve of methane production could be due to the ability of all tested pretreatments to degrade high strength organic compounds of landfill leachate and to decompose the nonbiodegradable organic macromolecules to biodegradable volatile fatty acids (VFA) with improved biodegradability [2]. Therefore, COD removal efficiencies were 46% and 48%, for FO and coagulation-flocculation pretreatments, respectively (Table 5). On the other hand, they contribute to reduce the NH_4-N content responsible for wastewater toxicity [43]. Therefore, the removal of free ammonia nitrogen (FAN), may avoid the bacterial inhibition and thus, the anaerobic digester failure [44]. It is clear that the air stripping pretreatment consistently increased methane production rate in comparison to all other pretreatments and led to a higher COD removal after anaerobic digestion (85%) (Table 6). For leachate, this pretreatment is usually used to eliminate NH_4-N [25,37] and it is also proposed as more robust and effective technique for enhancement of anaerobic digestion [45].

CONCLUSION

This study investigated the effect of coagulation–flocculation, Fenton oxidation and air stripping pretreatments on the improvement of LFL anaerobic digestion. For each pretreatment, the operational conditions are optimized.

In the case of coagulation-flocculation, a 46 and 18% of COD and ammonia removal were obtained respectively by adjusting the pH to 4 and $FeCl_3$ to 0.8 g/L. The BOD/COD ratio was increased from 0.3 to 0.37.

Under optimum conditions of FO (pH = 3, Fe^0=2 g/L and 3 g/L [H_2O_2]), the BOD/COD ratio increased from 0.3 to 0.39 and the ammonia removal was 23%.

Ammonia air stripping at its optimum operating conditions (pH =11, aeration rate =7 L/min and 18 h) achieved over 85% ammonia and 26% COD removal.

A comparison of three different pretreatment methods of LFL for biogas production was performed in batch condition. The methane production was improved for all the pretreated leachate when compared to the raw LFL.

After anaerobic digestion, the best methane yield using air stripping pretreatment was reached of 588 mL/g CODint.

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ABBREVIATIONS

AD	Anaerobic digestion
AOP	Advanced oxidation processes
BOD	Biological oxygen demand
BMP	Biochemical methane potential
C/N	Carbon to nitrogen ratio
CCD	Central Composite Design
COD	Chemical oxygen demand
CODint	Chemical oxygen demand introduced
FO	Fenton oxidation
LFL	Lanfill leachate
RSM	Response Surface Methodology
WWTP	Wastewater treatment plant
VS	Volatile solids
VSs to VSi	VS substrate to VS inocula

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